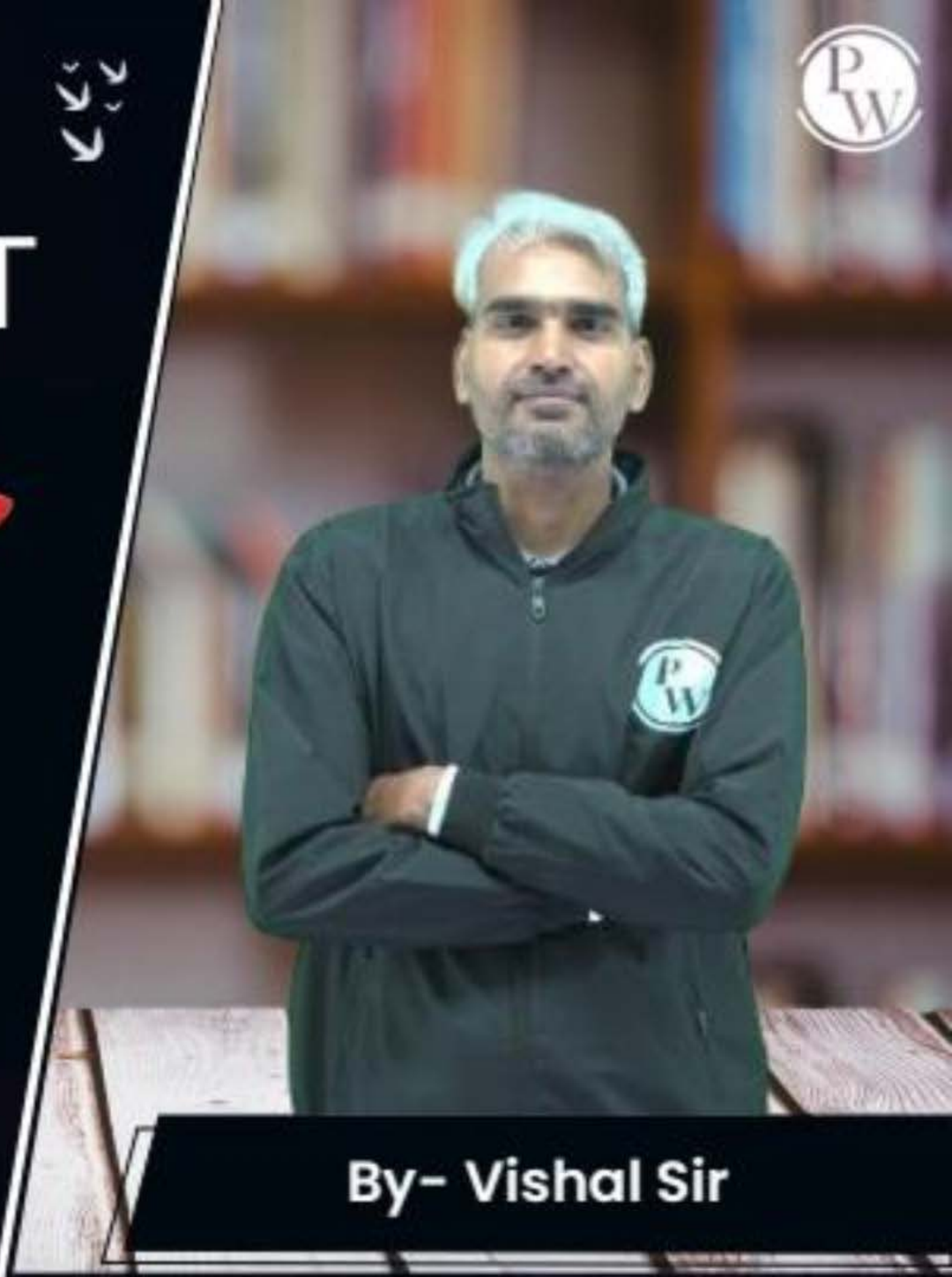
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 11



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Minimal cover or Canonical cover
- **Topic** Schema refinement (Normalization)
- **Topic** Problems because of redundancy in a relation

Slide

Topics to be Covered



Topic Properties of decomposition

Topic Dependency preserving decomposition



Topic : Schema refinement (Normalization)

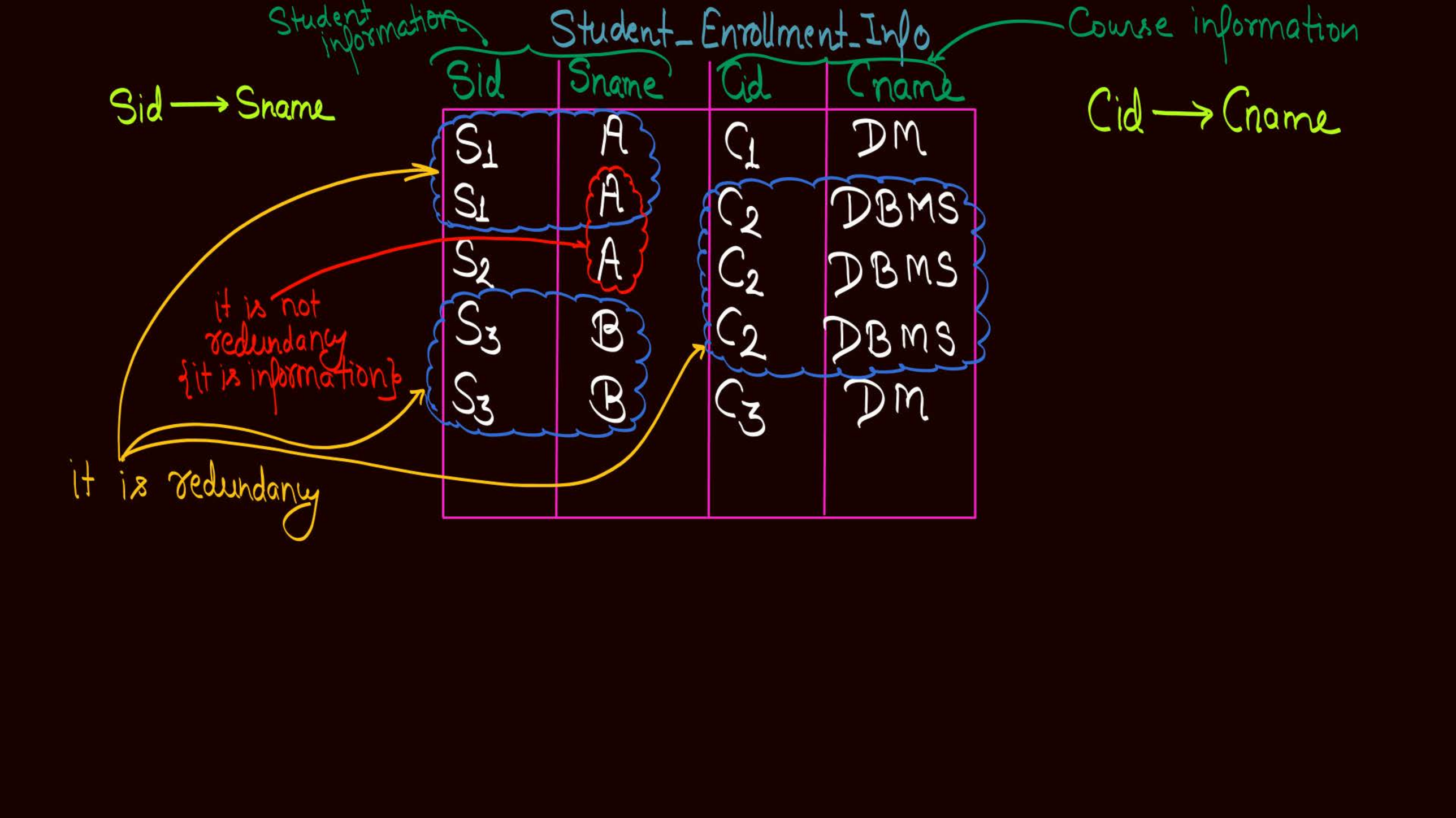
duplicate information

- Normalization is a process of reducing / eliminating the redundancy present in the relational table



Topic : Redundancy in a relation

If independent informations are stored in the same table, then redundancy is possible


Student information
Student_Enrollment_Info
Course information
Sid → Sname
Cid → Cname
Sid
Sname
Cid
Cname
S1
A
C1
DM
S1
A
C2
DBMS
S2
A
C2
DBMS
S3
B
C2
DBMS
S3
B
C3
DM
it is not redundancy {it is information}
it is redundancy

* Attributes of the primary key are denoted by "underlining" those attributes.

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

If {Sid, Cid} is the Primary key then Sid & Cid are not allowed to take "NULL" values

Let,

Sid → Sname
&
Cid → Cname

are only two FDs that holds in the relation.

∴ Candidate key of the relation is {Sid, Cid}

If we define a primary key for the relation then {Sid, Cid} will be Primary key

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

Sid → Sname

Cid → Cname



Topic : Problems because of redundancy

* If redundancy is present in the relation, then various problems are possible

* (1) It requires more storage space { it is not a very big Problem }

* (2) Insertion Anomaly
* (3) Deletion Anomaly
* (4) Updation Anomaly

Insertion Anomaly :-

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM
NULL	NULL	C_4	AI

Sid → Sname

Cid → Cname

But Sid Can not be NULL

∴ We can not insert (C_4, AI) in the relation.

Insertion Anomaly:- If independent informations are stored in the same relational table, then sometimes it may not be possible to insert one information without inserting other independent information.

eg: We can not insert the information of a new Course (C4, AI) until some students enroll for that Course, because "Sid" can not be NULL.

Deletion Anomaly :-

Suppose we want to delete information of student with Sid : S_1

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1 NULL	A NULL	C_1	DM
S_1 NULL	A NULL	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

Not allowed

∴ We will have to delete the Complete tuple

Sid → Sname

Cid → Cname

Deletion Anomaly :-

Suppose we want to delete information of Student with Sid : S_1

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

Sid → Sname

Cid → Cname

If we delete all the tuples w.r.t Student "S_1", then we will lose the information of Course "C_1" as well.

Deletion Anomaly :- Some times when we try to delete one independent information, we may lose some other independent information.

eg: If we try to delete the information of student "S_1" then information of Course with $Cid = C_1$ is lost.

Updation Anomaly :-

↓

Let us consider a situation in which we want to update the name of the Course with Cid = C_2 to DB & WH from DBMS

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	~~DBMS~~ DB & WH
S_2	A	C_2	~~DBMS~~ DB & WH
S_3	B	C_2	~~DBMS~~ DB & WH
S_3	B	C_3	DM

Sid → Sname

Cid → Cname

↳ Updation will be required in all duplicate Copies ⟹ ∴ This operation will be time Consuming

Updation Anomaly :-

Updation is required in all duplicate Copies, so it will be time consuming operation



Topic : Schema refinement (Normalization)

- Normalization is a process of decomposing (splitting) a relational tables into smaller tables (sub-relations) such that it eliminates/reduces the data redundancy, and it can overcome undesirable characteristics like Insertion, Updation and Deletion Anomalies

$F = \{ Sid \rightarrow Sname,\ Cid \rightarrow Cname \}$

Student_Enrollment_Info

Sid	Sname	Cid	Cname
S_1	A	C_1	DM
S_1	A	C_2	DBMS
S_2	A	C_2	DBMS
S_3	B	C_2	DBMS
S_3	B	C_3	DM

Assume that the relation is decomposed into following three sub-relations

Student

Sid	Sname
S_1	A
S_2	A
S_3	B

$F_1 = \{ Sid \rightarrow Sname \}$

C.K = {Sid}

Course

Cid	Cname
C_1	DM
C_2	DBMS
C_3	DM
C_4	AI

$F_2 = \{ Cid \rightarrow Cname \}$

C.K = {Cid}

Enroll

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2
S_3	C_2
S_3	C_3

$F_3 = \{$ Empty $\}$

C.K = {Sid, Cid}

After decomposition there is no redundancy & After decomposition all anomalies are overcome.



Topic : Properties of decomposition

While decomposing the relation into sub-relations Following must be ensured

① Decomposition must be dependency preserving decomposition:
i.e. All the functional dependencies present in the original relation must be preserved even after the decomposition into subrelations

i.e. After the decomposition there should not be any loss of information

neither in terms of functional dependencies

nor in terms of tuples present in the relation

② Decomposition must be lossless Join (⋈) decomposition:-
i.e. if we perform the natural join (⋈) of all the sub-relations, then we must get the exact same set of tuples as original relation

Properties of decomposition

1) Dependency Preserving decomposition

Let R be the relational schema with FD set F is decomposed into sub-relations $R_1, R_2, R_3 \ldots R_m$ with FD sets $F_1, F_2, F_3 \ldots F_m$ respectively

i.e. $R(F) \rightarrow R_1(F_1), R_2(F_2), R_3(F_3) \ldots R_m(F_m)$

then in general

$F_1 \cup F_2 \cup F_3 \cup \ldots \cup F_m \subseteq F$ { Always true }

if, $F_1 \cup F_2 \cup F_3 \cup \ldots \cup F_m = F$, then dependency Preserving

if, $F_1 \cup F_2 \cup F_3 \cup \ldots \cup F_m \subset F$, then not a dep. preserving decomposition

$F_1 \cup F_2 \cup F_3 \cup \ldots \cup F_m \supset F$ Not possible

2) Lossless Join decomposition

Let R be the relational schema with FD set F is decomposed into sub-relations $R_1, R_2, R_3 \ldots R_m$ with FD sets $F_1, F_2, F_3 \ldots F_m$ respectively

i.e. $R(F) \rightarrow R_1(F_1), R_2(F_2), R_3(F_3) \ldots R_m(F_m)$

then, in general,

$R_1 \bowtie R_2 \bowtie R_3 \bowtie \ldots \bowtie R_m \supseteq R$

if $=$, then lossless Join decomposition

if $\supset$, then lossy Join decomposition

if $\subset$, Not possible

Q :- Let $R(A, B, C, D)$ be the relational schema with FD set $F = \{A \rightarrow B,\ B \rightarrow C,\ C \rightarrow D,\ D \rightarrow A\}$

If relation R is decomposed into three sub-relations $R_1(A, B)$, $R_2(B, C)$, & $R_3(C, D)$, then identify whether the decompusition is dependency preserving decomposition or not ?

$R(A, B, C, D)$, $F = \{A \to B,\ B \to C,\ C \to D,\ D \to A\}$

∴ Decomposition is dependency Preserving decomposition

$R_1(A, B)$

$(A)^+ = \{A, B, C, D\}$

$(B)^+ = \{A, B, C, D\}$

$F_1 = \{A \to B,\ B \to A\}$

$R_2(B, C)$

$(B)^+ = \{B, C, D, A\}$

$(C)^+ = \{C, D, A, B\}$

$F_2 = \{B \to C,\ C \to B\}$

$R_3(C, D)$

$(C)^+ = \{C, D, A, B\}$

$(D)^+ = \{D, A, B, C\}$

$F_3 = \{C \to D,\ D \to C\}$

$F_1 \cup F_2 \cup F_3 = \{A \to B,\ B \to C,\ C \to D,\ B \to A,\ C \to B,\ D \to C\}$

① We already know $F_1 \cup F_2 \cup F_3 \subseteq F$ i.e. F covers $F_1 \cup F_2 \cup F_3$ → eqn ①

② We only need to check if $F_1 \cup F_2 \cup F_3$ covers F or not i.e. $F \subseteq F_1 \cup F_2 \cup F_3$ or not ?

FDs of F

$A \to B$ {directly present in $F_1 \cup F_2 \cup F_3$ ∴ Member}

$B \to C$ {directly present in $F_1 \cup F_2 \cup F_3$ ∴ Member}

$C \to D$ {directly present in $F_1 \cup F_2 \cup F_3$ ∴ Member}

$D \to A$, $(D)^+$ w.r.t. $F_1 \cup F_2 \cup F_3 = \{D, C, B, A\}$ $A \in (D)^+$ ∴ $D \to A$ is a member of $F_1 \cup F_2 \cup F_3$

All FDs of F are member of $F_1 \cup F_2 \cup F_3$

∴ $F \subseteq F_1 \cup F_2 \cup F_3$ — eqn ②

By ① & ②

$F_1 \cup F_2 \cup F_3 = F$

Q :- Let $R(A, B, C, D)$ be the relational schema with FD set $F = \{AB \rightarrow CD, D \rightarrow A\}$

If relation R is decomposed into two sub-relations $R_1(A, D)$ & $R_2(B, C, D)$, then identify whether the decomposition is dependency preserving decomposition or not ?

$R(A, B, C, D)$ $\quad F = \{ AB \rightarrow CD,\ D \rightarrow A \}$

$R_1(AD)$ $\quad F_1 = \{ D \rightarrow A \}$

$(A)^+ = \{A\}$

$(D)^+ = \{D, A\}$

$R_2(BCD)$ $\quad F_2 = \{ BD \rightarrow C \}$

$(B)^+ = \{B\}$

$(C)^+ = \{C\}$

$(D)^+ = \{D, A\}$

$(BC)^+ = \{B, C\}$

$(BD)^+ = \{B, D, A, C\}$

$(CD)^+ = \{C, D, A\}$

$F_1 \cup F_2 = \{ D \rightarrow A,\ BD \rightarrow C \}$

① We know $F_1 \cup F_2 \subseteq F$ — eqn ①

② Check if $F \subseteq F_1 \cup F_2$

FDs of F

$D \rightarrow A$ {Directly present in $F_1 \cup F_2$}

$AB \rightarrow CD \Rightarrow (AB)^+$ w.r.t. $F_1 \cup F_2 = \{A, B\} = CD \notin (AB)^+$

$\therefore AB \rightarrow CD$ is not a member of $F_1 \cup F_2$

Hence, $F \not\subseteq F_1 \cup F_2$ — eqn ②

By eqn ① & ② $F_1 \cup F_2 \subset F$ { Not a dependency preserving decomposition }

Q :- Let $R(A, B, C, D, E, F)$ be the relational schema with FD set $F = \{AB \to C,\ BC \to A,\ AC \to B,\ B \to D,\ AD \to E,\ E \to F\}$

Which of the following decomposition of R is dependency Preserving decomposition.

(1) $D_1 = \{R_1(ABC),\ R_2(ACDF),\ R_3(EF)\}$

(2) $D_2 = \{R_1(ABC),\ R_2(ABDE),\ R_3(EF)\}$

$F = \{ AB \to C,\ BC \to A,\ AC \to B,\ B \to D,\ AD \to E,\ E \to F \}$

(1) $D_1 = \{ R_1(ABC),\ R_2(ACDF),\ R_3(EF) \}$

$F_1 = \{ AB \to C,\ BC \to A,\ AC \to B \}$

$F_2 = \{ \}$

$F_3 = \{ E \to F \}$

$(B)^+ : \{B, D\}$

$B \to D$

$B \to$ Nothing in R_1

&

No other relation Contain B
$\therefore (B)^+$ will not be Computed in any other relation

$\therefore B \to D$ is not preserved after decomposition

$AD \to E$

(A,D) or F- not in R_1

$(AD)^+ = \{A, D, E, F\}$
$\therefore AD \to F$ in R_2

$(F)^+ = F$

$\therefore AD \to E$ is not Preserved after decomposition

$F = \{ AB \to C,\ BC \to A,\ AC \to B,\ B \to D,\ AD \to E,\ E \to F \}$

(2) $D_2 = \{ R_1(ABC),\ R_2(ABDE),\ R_3(EF) \}$

$F_1 = \{ AB \to C,\ BC \to A,\ AC \to B \}$ $F_2 = \{ B \to D,\ AD \to E \}$ $F_3 = \{ E \to F \}$

All FDs of F are preserved.

$$F = \{ AB \to C,\ BC \to A,\ AC \to B,\ B \to D,\ AD \to E,\ E \to F \}$$

(1) $D_1 = \{ R_1(ABC),\ R_2(ACDF),\ R_3(EF) \}$

$F_1 = \{ AB \to C,\ BC \to A,\ AC \to B \}$

$F_2 = \{ AC \to DF,\ AD \to F,\ ACD \to F \}$

$F_3 = \{ E \to F \}$

$B \to D$: $(B)^+$ w.r.t. $F_1 \cup F_2 \cup F_3 = \{B\}$, $D \notin B^+$ $\therefore$ $B \to D$ is lost

$AD \to E$: $(AD)^+$ w.r.t. $F_1 \cup F_2 \cup F_3 = \{A, D, F\}$, $E \notin (AD)^+$ $\therefore$ $AD \to E$ is lost

Lossless Join decomposition

H.W:- Read about following relational algebra operations.

1. Projection (π)
2. Selection (σ)
3. Cross Product ($\times$)
4. Natural Join ($\bowtie$)



2 mins Summary

Topic Properties of decomposition

Topic Dependency preserving decomposition

Slide

THANK - YOU